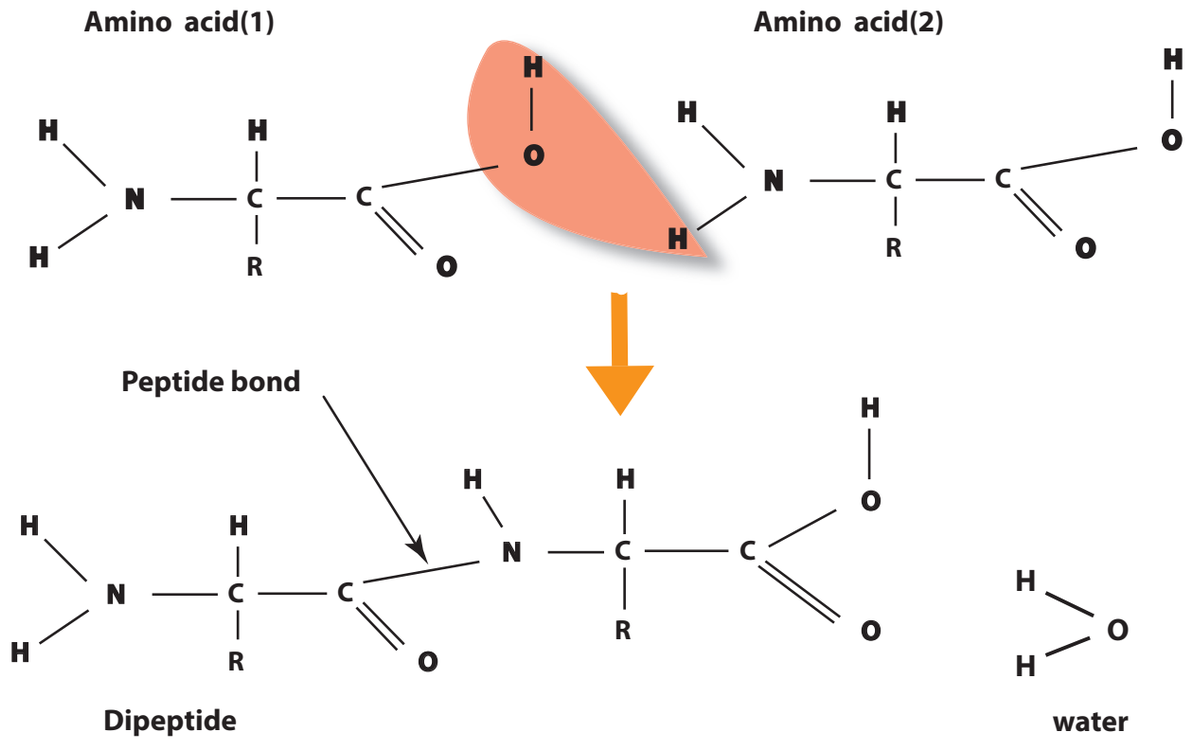


Fig 10.2: Structure of dipeptide

structure is known as dipeptide. Three amino acids bonded together to form a tripeptide. As additional amino acids join the chain, the structure becomes a polypeptide. Most proteins are polypeptides that are 100 to 300 amino acids long.

10.3 Classification of proteins

Proteins are large molecules formed by the combination of a number of aminoacids. About 20 amino acids have been found to occur in proteins and are important from the point of view of human nutrition. Amino acids can be classified as follows:

10.3.1 Nutritional Classification of Amino Acids

I. Essential Amino acids (Indispensable Amino acids)

An *essential* amino acid may be defined as one which is necessary for the growth and health of all living organisms and which cannot be synthesised in the body and must therefore be supplied through dietary intake. There are nine amino acids considered essential for the human infant, out of which Histidine is considered non-essential for the adult.

II. Semi-essential Amino acids (Conditionally Essential Amino acids)

Sometimes a non-essential amino acid can become essential. During illness or conditions of trauma, or in other special circumstances the need for an amino acid that is normally non-essential may become greater than the body's ability to produce it. In such circumstances, that amino acid becomes essential for the ill person. Amino acids that behave this way are referred to as **'Conditionally Essential'** amino acids for critically ill people.

Methionine can be converted to cystine, but cystine cannot be converted to

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